

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: ECC-207

Subject: Digital Logic & Computer Design

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Reduce the Boolean expression: $A + B'(A + C')$ [Note: Standard form reconstructed from original exam print; prime denotes logical complement].	3	CO1
(b)	What are don't care terms in Boolean logic?	3	CO1
(c)	Differentiate a decoder from a demultiplexer.	3	CO2
(d)	Draw the state table and excitation table of a T flip-flop.	3	CO3
(e)	Write about virtual memory and its architectural importance.	3	CO4
UNIT – I			
2. (a)	What is meant by duality in Boolean algebra? State and prove the Associative and Distributive theorems.	7	CO1
(b)	Minimize the expression $f(A, B, C, D) = \Sigma(0, 1, 2, 5, 8, 9, 10)$ using K-map and obtain: (i) Minimal SOP expression (ii) Minimal POS expression	8	CO1
3. (a)	Explain the operation of a magnitude comparator in detail.	7	CO2
(b)	Implement a full subtractor using an 8:1 multiplexer.	8	CO2
UNIT – II			
4. (a)	What is the Race-Around condition? How is it overcome in a JK flip-flop? Explain with a circuit/timing diagram.	5	CO3
(b)	Design a Mod-5 synchronous counter using JK flip-flops.	10	CO3
5. (a)	Explain the design procedure of converting a JK flip-flop into a T flip-flop.	7	CO3
(b)	Write short technical notes on: (i) Shift registers (ii) PLA & PAL architecture	8	CO2

[P.T.O.]

Q. No.	Question	Max. Marks	CO(s)
UNIT – III			
6. (a)	Differentiate between hardwired control and microprogrammed control. Explain the organization of a microprogrammed control unit with a neat block diagram.	10	CO4
(b)	What is an addressing mode? Explain the different types of addressing modes with suitable examples.	5	CO4
7. (a)	What are the different phases of an instruction cycle? Explain each phase in detail with the help of a flowchart.	7	CO4
(b)	What is assembly language? Explain the basic computer instruction formats with suitable diagrams.	8	CO4
UNIT – IV			
8. (a)	Define DMA (Direct Memory Access). Explain its need and operational working in detail with the help of a suitable block diagram.	7	CO4
(b)	What is an interrupt? Explain the complete interrupt cycle with the help of a flowchart.	8	CO4
9.	Write comprehensive technical notes on: a) Modes of data transfer (Programmed, Interrupt-driven, DMA) b) Cache memory architecture and mapping c) Associative memory (Content-Addressable Memory)	15 (5+5+5)	CO4

**** End of Question Paper ****